

Respiratory Isolation for Tuberculosis: A Historical Perspective

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Although tuberculosis is an ancient disease, recognition of its airborne route of transmission, with implications for respiratory isolation, is relatively recent. Historically, the disease was believed to be hereditary or contracted by inhaling ‘miasma,’ or corrupted air, and patients were not routinely isolated. The sanatorium initially served as a place of healing for patients with tuberculosis. Discovery of the tubercle bacillus and its mode of transmission via aerosol in the late 19th/early 20th century reinforced the important public health role of sanatoria and tuberculosis hospitals in preventing disease transmission through isolation. In recent decades, outpatients receiving treatment are required to remain in respiratory isolation at home until they are deemed noninfectious based on multiple sputum samples. This historical review demonstrates that despite changing medical knowledge, drug therapies, and social conditions over time, the role of isolation remains an important topic of debate in the treatment of patients with pulmonary tuberculosis.

Keywords. airborne; isolation practice; respiratory; transmission; tuberculosis.

This article is intended to provide a historical perspective on the conceptualization of tuberculosis and its epidemiology as the basis for evolving respiratory isolation practices, particularly in the context of recently revised public health guidelines for persons with pulmonary tuberculosis in community settings published by the National Tuberculosis Coalition of America (NTCA) [1].

CONCEPTIONS OF TUBERCULOSIS PATHOGENESIS IN ANTIQUITY THROUGH THE 20TH CENTURY

Mycobacterium tuberculosis has infected humans for thousands of years [2–4]. In the early Hippocratic corpus, a collection of chronic and debilitating respiratory illnesses, likely including tuberculosis, were referred to as phthisis, deriving from the Greek word φθίω, which means to waste away. The Hippocratic school considered phthisis to be hereditary [5], perhaps because it was often observed in cohabitating family members. Its potentially contagious nature was first recognized by Isocrates and

Aristotle, the latter of whom, in describing his observations of scrofula in domesticated pigs and oxen, commented that the illness could be transmitted by “foul air” [6, 7]. The physiologist and anatomist Galen also seemed to prefer the contagious theory of phthisis over the hereditary one, warning against intimate contact with sufferers of this malady [5]. However, physicians trained in the Galenic tradition preferred the theory that, as in the case of black death or cholera, a person became ill after inhaling *miasmata* (from the Greek word, μῑασμα, meaning foul air) emanating from putrefying carcasses, rotting vegetation, or other decomposing matter [8]. Therefore, individuals with phthisis were not ostracized during the classical Greek and Roman periods.

The Arabic physician and scholar Ibn Sina (Avicenna; 980–1037 CE) described phthisis in his *The Canon of Medicine* as a contagious disease and recommended the isolation of patients with overt pathology to limit its spread [9–11]. In his *De Contagione et Contagiosis Morbis* (*On Contagion and Contagious Diseases*), Girolamo Fracastoro (1478–1553 CE) advanced an early “germ theory” of disease, hypothesizing the transmissibility of various diseases, including tuberculosis, through tiny self-multiplying, seed-like bodies (*seminaria contagiorum*), which could be spread by direct or indirect contact through contaminated objects or passed through the air [12]. Fracastoro’s ideas had a discernible impact in Italy and beyond. The anatomist Antonio Valsalva and his pupil Giovanni Battista Morgagni refused to perform autopsies on persons who died of consumption for fear of contracting the disease [13]. Persons with scrofula, known as *morbus regius* or king’s evil in the Middle Ages because it was believed it could be cured by the

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touch of the English and French monarchs [14], began to be treated as lepers [15]. In the early 18th century, the Health Council of the Republic of Lucca imposed the isolation and treatment of patients suffering from consumption in distinct hospitals, thus laying the foundation of the sanatorium [10].

Notwithstanding the epidemiological observations of several scholars, such as Benjamin Martin, who in 1720 theorized that consumption, like the plague, could be transmitted between humans through “some certain species of *animalcula*” [16], the prevailing school of thought in northern Europe until the mid-19th century was that the disease was hereditary [17] or that it was due to the inhalation of miasma [18]. Undoubtedly inspired by Louis Pasteur’s groundbreaking discoveries on the microbial origins of disease, the French military physician Jean Antoine Villemin presented to the Academy of Medicine in December 1865 a series of experiments in which he demonstrated that tuberculous lymph node tissues, when inoculated into rabbits, uniformly resulted in typical disseminated tubercles, while noninoculated rabbits remained disease free [19]. Although he was able to show conclusively the infectiousness of the bronchial secretions of diseased animals in his *Etudes sur la Tuberculose* in 1868 [20], his findings were met with a mixture of skepticism and lukewarm interest, because they did not include microscopic identification or isolation of the etiological infectious agent.

In a brief yet landmark presentation to the Berlin Physiological Society on 24 March 1882, the then 39-year-old Robert Koch provided incontrovertible proof of the microbial basis of consumption [21]. Through rigorous experiments employing various animal models and routes of inoculation, and which were expressed in his now-famous postulates, Koch identified the tubercle bacillus (subsequently classified as *Mycobacterium tuberculosis*), which was both necessary and sufficient to cause the disease. In his concluding comments to his audience, he also noted that the bacillus could be transmitted directly from human to human without an external, natural reservoir; the portal of entry was the lungs; and that persons with cavitary disease could expectorate the bacilli into the air and these could be inhaled by others [7]. Paul Ehrlich, who attended the lecture, refined Koch’s bacillary staining technique, which was further modified by Ziehl and Neelsen and which remains to this day as the main method for diagnosing tuberculosis in endemic areas of the world through the detection of acid-fast bacilli (AFB) in smears prepared from patient sputum samples [22].

Despite Koch’s persuasive arguments, many physicians and public health officials were slow to accept that consumption could be transmitted between individuals via the airborne route, and respiratory isolation was not considered practical. Although the influential public health officer and germ theory proponent Charles Chapin led several community hygiene practices to limit pandemic flu and other communicable diseases in Providence, RI, he argued that “no one advocates the isolation... of pulmonary tuberculosis... simply because infected persons are so

numerous that it is impossible” [23]. At the turn of the 20th century, Koch’s colleague Carl Flügge proposed that the tubercle bacillus was transmitted to new hosts by fine respiratory droplets, demonstrating that exposure to such droplets expectorated from patients with tuberculosis could lead to infection of guinea pigs [24]. The theory that tuberculosis was transmitted by Flügge’s droplets prevailed until the 1930s, when William Firth Wells first differentiated between respiratory droplets of varying diameters, advancing instead the concept of droplet nuclei, which resulted from desiccation of fine droplets, and which could harbor infectious bacilli as they drifted in the air for many minutes or even hours “like cigarette smoke” [25].

Wells and his colleague Richard L. Riley tested the validity of the droplet-nucleus concept in the Veterans Administration Hospital in Baltimore, MD in the late 1950s. A pilot study ward contained 6 negative-pressure rooms whose effluent air was either left unperturbed or UV-irradiated and then channeled to a rooftop facility containing cages of tuberculosis-naïve guinea pigs. Based on the drug-susceptibility profiles and temporal patterns, new guinea pig infections (as measured by tuberculin skin test conversion) could be traced back to individual patients, and other potential confounding sources of infection (eg, animal handlers) could be excluded by the absence of infections among animals exposed to UV-irradiated effluent air. These series of experiments allowed Wells and Riley to make several important conclusions, including (1) one infectious dose (or “quantum”) was present per 11 000–12 500 cubic feet of air; (2) not all patients were equally infectious (eg, 1 patient with laryngeal tuberculosis accounted for 46% of new guinea pig infections, despite residing on the ward for only 1% of the total patient days); and (3) untreated patients were up to 50-fold more likely to be infectious compared to those who had initiated tuberculosis treatment [26–28].

Applying powdered stains to the oral surfaces of subjects and using closely held glass slides to capture drop marks and measure their diameters, research teams in the early 20th century found that certain respiratory maneuvers, such as speaking loudly, coughing, and sneezing, could generate a range of respiratory droplet numbers and sizes, which were a function of the viscosity of the expelled liquid and the velocity of the air stream generated. The Scottish physician and bacteriologist James P. Duguid determined that droplet nuclei produced by sneezing, coughing, or speaking had a diameter of < 12 µm (similar to those generated by artificial air-stream atomization), and could harbor pathogenic bacteria suspended in the air for an average of 30–60 minutes and up to 30 hours [29]. Thus, although for many years it had been known that tuberculosis infection and disease resulted from prolonged and intimate contact with coughing source patients, the findings by Wells and Duguid suggested that the infectious risk had less to do with the total number of bacilli expelled by a patient than with the concentration of infectious droplet nuclei, that is, those suitable for deposition in the alveoli, in relationship to the volume of

air inhaled by the individual's contacts [7]. Importantly, the droplet-nucleus concept of *M. tuberculosis* airborne transmission was highly useful in explaining the specific transmission patterns observed in various tuberculosis outbreaks, such as the one at the Arkansas Negro Boys Industrial School [30] and aboard the US navy ship, USS *Richard E. Byrd* [31].

THE EMERGENCE OF SANATORIA

In the second century BCE, Galen recommended fresh air, milk, and sea voyages for the treatment of phthisis, and, as stated above, individuals with scrofula were treated with the royal touch throughout the middle ages [18]. However, from the end of the 18th century, physicians began to recommend that patients with advanced disease should take their medicines and follow advice about food, physical exercise, and fresh air whilst living at home, rather than exerting themselves by travelling either to drier climes, invigorating coastal resorts, or spas [32].

Before the popularization of sanatoria towards the end of the 19th century, the admission of patients with tuberculosis to general hospitals was discouraged; consumption—with a prolonged period of sickness, uncertain prospect of recovery, and a frequently fatal outcome—resisted the notion that hospitals were curative institutions deserving of public support. Nevertheless, some patients were hospitalized and separated from other patients, although this often proved difficult [33]. This was in line with the broader approach of pavilion design for general hospitals and fever hospitals that used spatial segregation and ventilation systems to address cross-infection [33]. A far greater number of afflicted patients found themselves not in hospitals but in almshouses and workhouses where, by and large, such separation was much less likely [34].



Figure 1. Edward Trudeau opened America's first sanatorium at Saranac Lake, NY in the 1880s. Patients sat on the wide sun porches to take in the fresh air cure.

The first tuberculosis sanatorium was founded in the Silesian Mountain village of Gobersdorf, Germany in 1859 by Herman Brehmer. It is important to note that sanatoria first emerged as a treatment, rather than public health, modality for patients with consumption. In addition to the concept that fresh mountain air was therapeutic for this disease, Brehmer and others believed that patients with consumption had small hearts relative to the sizes of their lungs and reasoned that the lower oxygen content of inspired air at high altitude would lead to cardiac enlargement and hence improvement in their health [35]. Edward Livingston Trudeau established the first US sanatorium at Saranac Lake in the Adirondack mountains (Figure 1), and the first 2 patients were admitted to the original "Little Red" cottage in February 1884 (Figure 2). Trudeau, who himself was diagnosed with tuberculosis in the early 1870s and lived until 1915, attributed his longevity to the salutary effects of this Adirondacks retreat [7].

Acknowledgment that tuberculosis was a transmissible bacterial disease gave a fillip to the sanatorium movement towards the end of the 19th century, because germ theory provided a scientific justification for isolating patients. Further stimulus came from the provision of sanatorium benefits through health insurance schemes. Absent therapeutic successes—the dashed hopes for Robert Koch's tuberculin treatment further emphasized a need for sanatoria [36]—diet, rest therapy, graduated exercise, and abundant fresh air were the central pillars of the tuberculosis sanatorium regimen [37]. This regimen was scaffolded through close medical surveillance on the one hand and patient self-management on the other hand. Regimens differed from place to place, but in general resident patients received education on



Figure 2. "Little Red," the original cure cottage of the Adirondack Cottage Sanitarium, built in 1884.

personal and domestic hygiene, safe expectorate or sputum disposal, and appropriate interactions with others [38].

Although the design of sanatoria varied widely, emphasis on the isolation of individual patients was more pronounced than in general medical facilities [39]. Brehmer's sanatorium, although based on a pavilion plan, accommodated patients in bedrooms rather than wards. Individual sleeping spaces at the Adirondack Cottage Sanatorium, Saranac Lake, NY similarly aimed to prevent cross-infection [40]. In many sanatoria individual patient rooms contained a wash basin, hot and cold running water, and a radiator, to minimize the use of common facilities (although socializing was permitted) [41]. Pivoting glass doors opened onto balconies. Indeed, the use of glass was such that the Zonnestraal Sanatorium in the Netherlands has been described as "a machine for light and air" [42]. Nevertheless, as evidenced by contemporaneous literature, isolation practices were hardly stringent, and visitors freely interacted with inpatients with apparently little concern for contagion [43].

The number of sanatorium beds in the United States rose from 11 952 in 1908 to 97 720 in 1942 [7]. Despite their proliferation, the curative benefits of sanatoria were openly questioned [44]. In the case of municipal sanatoria, many sufferers were confined involuntarily and the aspirational isolation of patients was far from the reality [40, 45]. Moreover, the vast majority of persons suffering from tuberculosis never saw the inside of a sanatorium [46]. Rather, they were encouraged to adopt the sanatorium regimen in their homes. There was a steady market for manuals authored by medical professionals on how patients should live with tuberculosis in community settings [47]. The use of well-ventilated revolving huts in yards and gardens (Figure 3), as well as the construction of screened, fresh-air porches attached to homes, are examples of how patients adapted the experience of isolation in the sanatorium to domestic circumstances [48, 49].

THE DISCOVERY OF ANTIBIOTICS FOR TUBERCULOSIS AND THE END OF THE SANATORIA

Patients continued to be hospitalized in sanatoria through the mid-20th century [37], when the treatment of tuberculosis was suddenly transformed with the discovery of several antibiotics with antitubercular activity. Working together with the Swedish pharmaceutical firm Ferrosan, Jörgen Lehmann showed in 1941 that the compound para-aminosalicylate (PAS) could potently inhibit *M. bovis* bacille Calmette-Guérin (BCG) and protect guinea pigs infected with virulent tubercle bacilli [50]. In the same year, Selman Waksman and Albert Schatz at Rutgers University isolated from the soil organism *Streptomyces griseus* an antibiotic they termed streptomycin, which inhibited the growth of virulent *M. tuberculosis* and could be used safely to treat *M. tuberculosis*-infected guinea pigs and patients with tuberculosis [51, 52].

Although the discovery of streptomycin revolutionized tuberculosis treatment, its intravenous formulation required daily clinic



Figure 3. Outdoor isolation unit for patient with tuberculosis in the backyard of her family's home on La Delta project, Thomastown, Louisiana.

visits and patients often relapsed due to the emergence of streptomycin-resistant strains. Indeed, some sanatoria remained part of the patient experience of tuberculosis into the 1960s as they were used to counter the possibility of relapse [53]. With the discovery of isonicotinic acid hydrazide (isoniazid) in 1952, nearly universal, life-long cures were possible for patients with tuberculosis receiving this drug in combination with streptomycin and PAS. Finally, the discovery of rifampicin in the mid-1960s and the recognition of pyrazinamide's sterilizing activity in the early 1970s allowed for a fully oral, 6-month regimen, which remains the current short-course therapy for drug-susceptible tuberculosis. To ensure the type of medical adherence otherwise possible only in hospital and to prevent ongoing transmission in the community setting, so-called directly-observed therapy (DOT) with intermittent drug dosing continues to be implemented to this day in outpatient clinics throughout the world [54, 55].

MODERN PRACTICES OF RESPIRATORY ISOLATION FOR PATIENTS WITH PULMONARY TUBERCULOSIS

An influential review article published in 1976 helped to establish the "2-week rule" of respiratory isolation following tuberculosis treatment initiation for patients with pulmonary disease and AFB-positive sputum [56]. This recommendation, which guided infection control practices for nearly 2 decades, was based on several epidemiological studies showing that *M. tuberculosis* transmission among household contacts was strongly correlated with the number of sputum AFB in the index cases, as well as the observation that tuberculosis chemotherapy usually rendered patients noninfectious within 2 weeks [57].

Subsequent US public health guidelines placed primacy on the AFB status of patients undergoing tuberculosis treatment, recommending that respiratory isolation, either in the hospital

or at home, be continued until at least 3 sputum samples collected over a 1 to 3-day period are AFB negative by smear microscopy [58]. Under home isolation, the patient's activities are restricted to the home, they are instructed to avoid social gatherings, and children under the age of 5 years are advised to leave the home until the patient is no longer deemed infectious [59]. Most state and local jurisdictions have the legal authority to enforce these guidelines under regulations pertaining to control of communicable diseases. Because patients may continue to have AFB-positive sputum for many months, examples abound of patients who have had to endure personal and economic hardships, including being separated from small children or having to quit school or their jobs, until they are released from home isolation [60].

However, substantial evidence [1] suggests that tuberculosis treatment rapidly renders patients noninfectious, including those with multidrug-resistant tuberculosis [61], even if their sputum smears remain AFB-positive, arguing against their respiratory isolation. A 1966 study conducted in Madras, India found that tuberculosis attack rates over a 5-year follow-up period were equivalent among close family contacts of all patients treated with isoniazid and PAS for 1 year, half of whom were randomized to treatment in the sanatorium and the other half to treatment at home [62]. More recently, studies have confirmed that the infectiousness of tuberculosis cases correlates more closely with the ability to generate aerosols of virulent *M. tuberculosis*-bearing droplet nuclei rather than with sputum bacillary load, as measured by acid-fast staining or microbiological culture, or intimacy/frequency of contact [63–65]. The rapid decline in infectiousness following effective tuberculosis treatment may also be attributed to antibiotic-induced changes in the transcriptional and metabolic profile of *M. tuberculosis* in sputum [66, 67].

CONCLUSIONS

“Isolation” has multiple meanings in the history of tuberculosis treatment and care. In one sense, it has involved patients seeking, or being forcibly admitted to, institutions in remote areas away from population centers. In another, it has meant implementing some form of precautions, including physical separation, in both clinical and domestic settings for an indeterminate amount of time, depending on the course of the disease in an individual patient. The inadequacy of the sputum smear AFB status in dictating respiratory isolation recommendations for patients on tuberculosis treatment, as well as the economic, social, and mental health consequences faced by such patients due to unnecessarily prolonged respiratory isolation, have provided the impetus for proposing a revision in the current guidelines [1]. As this review indicates, respiratory isolation practices for patients receiving treatment for pulmonary tuberculosis likely will continue to evolve in the future as additional evidence accumulates.

Notes

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